Appendix

Appendix A. Derivation of the ℓ^I-Based CSPA Variants

Appendix A.1. Derivation of the CSPA-ℓ^I Updates

The CSPA-ℓ^I algorithm is derived from the following constrained optimization problem, which seeks the minimal change to the weight vector ω_t such that the loss is zero.

$$\omega_{t+1} = \arg\min_{\omega \in \mathbb{R}^m} \frac{1}{2}\|\omega - \omega_t\|^2 \quad \text{subject to} \quad \ell^I(\omega; (\mathbf{x}_t, y_t)) = 0, \tag{A.1}$$

where the loss function ℓ^I is defined as:

$$\ell^I(\omega; (\mathbf{x}_t, y_t)) = \max\left(0, \left(\rho * \mathbb{I}_{(y_t=1)} + \mathbb{I}_{(y_t=-1)}\right) - y_t\left(\omega \cdot \mathbf{x}_t\right)\right), \tag{A.2}$$

To solve this problem, we introduce the Lagrangian with a Lagrange multiplier τ_t:

$$\mathcal{L}(\omega, \tau_t) = \frac{1}{2}\|\omega - \omega_t\|^2 + \tau_t\left(\left(\rho * \mathbb{I}_{(y_t=1)} + \mathbb{I}_{(y_t=-1)}\right) - y_t(\omega \cdot \mathbf{x}_t)\right), \tag{A.3}$$

The optimality condition is found by taking the partial derivative of the Lagrangian with respect to ω and setting it to zero:

$$\frac{\partial \mathcal{L}}{\partial \omega} = \omega - \omega_t - \tau_t y_t \mathbf{x}_t = 0, \tag{A.4}$$

This yields the formula for the new weight vector, ω:

$$\omega = \omega_t + \tau_t y_t \mathbf{x}_t, \tag{A.5}$$

By substituting this back into the constraint $\ell^I(\omega_{t+1}; (\mathbf{x}_t, y_t)) = 0$, we can solve for the multiplier τ_t.

$$\left(\rho * \mathbb{I}_{(y_t=1)} + \mathbb{I}_{(y_t=-1)}\right) - y_t((\omega_t + \tau_t y_t \mathbf{x}_t) \cdot \mathbf{x}_t) = 0, \tag{A.6}$$

Rearranging the terms to solve for τ_t gives:

$$\left(\rho * \mathbb{I}_{(y_t=1)} + \mathbb{I}_{(y_t=-1)}\right) - y_t(\omega_t \cdot \mathbf{x}_t) = \tau_t y_t^2(\mathbf{x}_t \cdot \mathbf{x}_t) = \tau_t\|\mathbf{x}_t\|^2, \tag{A.7}$$

Thus, the step size τ_t is determined by the hinge loss at time t:

$$\tau_t = \frac{\left(\rho * \mathbb{I}_{(y_t=1)} + \mathbb{I}_{(y_t=-1)}\right) - y_t(\omega_t \cdot \mathbf{x}_t)}{\|\mathbf{x}_t\|^2} = \frac{\ell^I(\omega_t; (\mathbf{x}_t, y_t))}{\|\mathbf{x}_t\|^2}. \tag{A.8}$$

Appendix A.2. Derivation of the CSPA$_1$-ℓ^I Updates

The CSPA$_1$-ℓ^I variant introduces a slack variable ξ and a regularization parameter C_t, modifying the objective function as follows:

$$\omega_{t+1} = \arg\min_{\omega \in \mathbb{R}^m, \xi \geq 0} \frac{1}{2}\|\omega - \omega_t\|^2 + C_t\xi \quad \text{subject to} \quad \ell^I(\omega; (\mathbf{x}_t, y_t)) \leq \xi, \tag{A.9}$$

The Lagrangian corresponding to the above optimization problem is given by:

$$\mathcal{L}(\omega, \xi, \tau_t, \kappa) = \frac{1}{2}\|\omega - \omega_t\|^2 + C_t\xi + \tau_t\left[\ell^I(\omega; (\mathbf{x}_t, y_t)) - \xi\right] - \kappa(\xi), \tag{A.10}$$

where $\tau \geq 0$ and $\kappa \geq 0$ are Lagrange multipliers. The optimality conditions are derived by differentiating $\mathcal{L}$ with respect to ω and ξ and setting the results to zero.

$$\frac{\partial \mathcal{L}}{\partial \omega} = \omega - \omega_t - \tau_t y_t \mathbf{x}_t = 0 \implies \omega = \omega_t + \tau_t y_t \mathbf{x}_t \tag{A.11}$$

$$\frac{\partial \mathcal{L}}{\partial \xi} = C_t - \tau_t - \kappa = 0 \implies \kappa = C_t - \tau_t \tag{A.12}$$

From the Karush-Kuhn-Tucker (KKT) conditions, we have $\tau_t \geq 0$ and $\kappa = C_t - \tau_t \geq 0$, which implies that $0 \leq \tau_t \leq C_t$. We now study two possible cases. In the first case, when $\ell^I(\omega_t; (\mathbf{x}_t, y_t))/\|\mathbf{x}_t\|^2 \leq C_t$. From this, one can obtain the derivation identically to the standard PA update, leading to the closed-form solution $\tau_t = \ell^I(\omega_t; (\mathbf{x}_t, y_t))/\|\mathbf{x}_t\|^2$. In the second case, where $\ell^I(\omega_t; (\mathbf{x}_t, y_t))/\|\mathbf{x}_t\|^2 > C_t$, this inequality can be equivalently denoted as:

$$C_t\|\mathbf{x}_t\|^2 < 1 - \max\left(0, \left(\rho * \mathbb{I}_{(y_t=1)} + \mathbb{I}_{(y_t=-1)}\right) - y_t\left(\omega \cdot \mathbf{x}_t\right)\right). \tag{A.13}$$

We also know from the constraint in Eq. (A.9) should satisfy at the optimum, so:

$$\max\left(0, \left(\rho * \mathbb{I}_{(y_t=1)} + \mathbb{I}_{(y_t=-1)}\right) - y_t\left(\omega \cdot \mathbf{x}_t\right)\right) \leq \xi.$$

Substituting the explicit form of ω from Eq. (A.11) into the constraint gives:

$$\max\left(0, \left(\rho * \mathbb{I}_{(y_t=1)} + \mathbb{I}_{(y_t=-1)}\right) - y_t\left(\omega_t \cdot \mathbf{x}_t\right)\right) - \tau\|\mathbf{x}_t\|^2 \leq \xi. \tag{A.14}$$

Combining this equation with Eq. (A.13) leads to:

$$(C_t - \tau)\|\mathbf{x}_t\|^2 < \xi.$$

Given that $\tau \leq C_t$, it follows that $\xi > 0$. Applying the KKT complementarity condition $\xi\kappa = 0$, and knowing that $\xi > 0$, we conclude that $\kappa = 0$. Substituting $\kappa = 0$ into Eq. (A.12) results in $\tau = C_t$. In summary, when $\ell^I(\omega_t; (\mathbf{x}_t, y_t))/\|\mathbf{x}_t\|^2 > C_t$, the KKT conditions imply that the optimal update is $\tau = C_t$. Taking all cases into account, we define τ_t as:

$$\tau_t = \min\left(C_t, \frac{\ell^I(\omega_t; (\mathbf{x}_t, y_t))}{\|\mathbf{x}_t\|^2}\right), \tag{A.15}$$

Thus, the final weight update rule is given by:

$$\omega_{t+1} = \omega_t + \tau_t y_t \mathbf{x}_t. \tag{A.16}$$

Appendix A.3. Derivation of the CSPA$_2$-ℓ^I Updates

The CSPA$_2$-ℓ^I is the following optimization problem:

$$\omega_{t+1} = \arg \min_{\omega \in \mathbb{R}^m, \xi \geq 0} \frac{1}{2}\|\omega - \omega_t\|^2 + C_t \xi^2 \quad \text{subject to} \quad \ell^I(\omega; (\mathbf{x}_t, y_t)) \leq \xi, \tag{A.17}$$

By substituting the ℓ^I formula in the constraint of the above optimization problem, we have $\left(\rho * \mathbb{I}_{(y_t=1)} + \mathbb{I}_{(y_t=-1)}\right) - y_t(\omega \cdot \mathbf{x}_t) \leq \xi$. Then, the Lagrangian of the optimization problem (A.16) equals:

$$\mathcal{L}(\omega, \xi, \tau_t) = \frac{1}{2}\|\omega - \omega_t\|^2 + C_t \xi^2 + \tau_t \left[\left(\rho * \mathbb{I}_{(y_t=1)} + \mathbb{I}_{(y_t=-1)}\right) - y_t(\omega \cdot \mathbf{x}_t) - \xi\right], \tag{A.18}$$

where $\tau_t \geq 0$ is a Lagrange multiplier. The optimality conditions are obtained by minimizing the Lagrangian with respect to ω and ξ:

$$\frac{\partial \mathcal{L}}{\partial \omega} = \omega - \omega_t - \tau_t y_t \mathbf{x}_t = 0 \implies \omega = \omega_t + \tau_t y_t \mathbf{x}_t \tag{A.19}$$

$$\frac{\partial \mathcal{L}}{\partial \xi} = 2C_t \xi - \tau_t = 0 \implies \xi = \frac{\tau_t}{2C_t} \tag{A.20}$$

Substituting the expressions for ω and ξ into the Lagrangian function and setting the partial derivative with respect to τ to zero will give:

$$-\tau_t \|\mathbf{x}_t\|^2 - \frac{\tau_t}{2C_t} + \left[\left(\rho * \mathbb{I}_{(y_t=1)} + \mathbb{I}_{(y_t=-1)}\right) - y_t(\omega_t \cdot \mathbf{x}_t)\right] = 0, \tag{A.21}$$

Rearranging and solving for τ_t yields:

$$\tau_t \left(\|\mathbf{x}_t\|^2 + \frac{1}{2C_t}\right) = \left(\rho * \mathbb{I}_{(y_t=1)} + \mathbb{I}_{(y_t=-1)}\right) - y_t(\omega_t \cdot \mathbf{x}_t), \tag{A.22}$$

Thus, the update parameter τ_t is given by:

$$\tau_t = \frac{\left(\rho * \mathbb{I}_{(y_t=1)} + \mathbb{I}_{(y_t=-1)}\right) - y_t(\omega_t \cdot \mathbf{x}_t)}{\|\mathbf{x}_t\|^2 + \frac{1}{2C_t}} = \frac{\ell^I(\omega_t; (\mathbf{x}_t, y_t))}{\|\mathbf{x}_t\|^2 + \frac{1}{2C_t}}. \tag{A.23}$$

Appendix B. Derivation of the ℓ^{II}-Based CSPA Variants

Appendix B.1. Derivation of the CSPA-ℓ^{II} Updates

The CSPA-ℓ^{II} algorithm is derived by solving the following constrained optimization problem:

$$\omega_{t+1} = \arg \min_{\omega \in \mathbb{R}^m} \frac{1}{2}\|\omega - \omega_t\|^2 \quad \text{subject to} \quad \ell^{II}(\omega; (\mathbf{x}_t, y_t)) = 0, \tag{B.1}$$

The corresponding Lagrangian for the above optimization problem is given by:

$$\mathcal{L}(\omega, \tau_t) = \frac{1}{2}\|\omega - \omega_t\|^2 + \tau_t \left[\left(\rho * \mathbb{I}_{(y_t=1)} + \mathbb{I}_{(y_t=-1)}\right) * \max(0, 1 - y_t(\omega \cdot \mathbf{x}_t))\right], \tag{B.2}$$

The optimality condition is obtained by differentiating $\mathcal{L}$ with respect to ω and setting it to zero:

$$\frac{\partial \mathcal{L}}{\partial \omega} = \omega - \omega_t - \tau_t \left(\rho * \mathbb{I}_{(y_t=1)} + \mathbb{I}_{(y_t=-1)}\right) y_t \mathbf{x}_t = 0, \tag{B.3}$$

This provides the update rule for the weight vector in terms of τ_t:

$$\omega = \omega_t + \tau_t \left(\rho * \mathbb{I}_{(y_t=1)} + \mathbb{I}_{(y_t=-1)}\right) y_t \mathbf{x}_t, \tag{B.4}$$

Substituting Eq. (B.4) into the Lagrangian yields:

$$\begin{aligned}\mathcal{L}(\tau_t) = &\frac{1}{2}\tau_t^2(\rho \cdot I(y_t=1) + I(y_t=-1))^2\|\mathbf{x}_t\|^2 \\ &+ \tau_t\Bigg[\left(\rho * \mathbb{I}_{(y_t=1)} + \mathbb{I}_{(y_t=-1)}\right) \cdot \max\Big(0, 1 - y_t(\omega_t \cdot \mathbf{x}_t) \\ &- \tau_t(\rho \cdot I(y_t=1) + I(y_t=-1))\|\mathbf{x}_t\|^2\Big)\Bigg],\end{aligned}$$

Next, we differentiate the expression with respect to τ_t.

$$\frac{d\mathcal{L}}{d\tau_t} = -\tau_t \left(\rho * \mathbb{I}_{(y_t=1)} + \mathbb{I}_{(y_t=-1)}\right)^2 \|\mathbf{x}_t\|^2 + (\rho * \mathbb{I}_{(y_t=1)} + \mathbb{I}_{(y_t=-1)}(1 - y_t(\omega_t \cdot \mathbf{x}_t)),$$

$$\tau_t = \frac{1 - y_t(\omega_t \cdot \mathbf{x}_t)}{\left(\rho * \mathbb{I}_{(y_t=1)} + \mathbb{I}_{(y_t=-1)}\right) \|\mathbf{x}_t\|^2}. \tag{B.5}$$

Appendix B.2. Derivation of the CSPA$_1$-ℓ^{II} Updates

CSPA$_1$-ℓ^{II} extends the CSPA$_1$ using ℓ^{II} loss function using the following optimization problem:

$$\omega_{t+1} = \arg\min_{\omega\in\mathbb{R}^m, \xi\geq 0} \frac{1}{2}\|\omega - \omega_t\|^2 + C_t\xi \quad \text{subject to} \quad \ell^{II}(\omega; (\mathbf{x}_t, y_t)) \leq \xi, \tag{B.6}$$

The Lagrangian for this problem is formulated as:

$$\mathcal{L}(\omega, \xi, \tau_t, \lambda) = \frac{1}{2}\|\omega - \omega_t\|^2 + C_t\xi + \tau_t\left[(\rho * \mathbb{I}_{(y_t=1)} + \mathbb{I}_{(y_t=-1)}) * \max(0, 1 - y_t(\omega \cdot \mathbf{x}_t)) - \xi\right] - \kappa\xi, \tag{B.7}$$

where $\tau \geq 0$ and $\kappa \geq 0$ are Lagrange multipliers. The optimality conditions are found by differentiating $\mathcal{L}$ with respect to ω and ξ. Therefore, we have:

$$\frac{\partial \mathcal{L}}{\partial \omega} = \omega - \omega_t - \tau_t \left(\rho * \mathbb{I}_{(y_t=1)} + \mathbb{I}_{(y_t=-1)}\right) y_t \mathbf{x}_t = 0 \tag{B.8}$$

$$\frac{\partial \mathcal{L}}{\partial \xi} = C_t - \tau_t - \kappa = 0. \tag{B.9}$$

By rearranging Eqs. (B.8) and (B.9) , we obtain:

$$\omega = \omega_t + \tau_t(\rho \cdot I_{(y_t=1)} + I_{(y_t=-1)})y_t\mathbf{x}_t, \tag{B.10}$$

$$\kappa = C_t - \tau_t, \tag{B.11}$$

Substituting these expressions back into the Lagrangian, differentiating the resulting function with respect to τ_t, and applying the KKT condition $0 \leq \tau_t \leq C_t$, yields the update parameter:

$$\tau_t = \min\left\{ C_t, \frac{\ell^{II}(\omega_t; (\mathbf{x}_t, y_t))}{\left(\rho * \mathbb{I}_{(y_t=1)} + \mathbb{I}_{(y_t=-1)}\right)^2 \|\mathbf{x}_t\|^2} \right\}. \tag{B.12}$$

where $\ell^{II}(\omega_t; (\mathbf{x}_t, y_t))$ is the loss incurred by the current weight vector ω_t.

Appendix B.3. Derivation of the CSPA$_2$-ℓ^{II} Updates

CSPA$_2$-ℓ^{II} employs the ℓ^{II} loss function within the CSPA$_2$ framework, as specified in Eq. 13, and is given by:

$$\omega_{t+1} = \arg \min_{\omega \in \mathbb{R}^m, \xi \geq 0} \frac{1}{2}\|\omega - \omega_t\|^2 + C_t\xi^2 \quad \text{subject to} \quad \ell^{II}(\omega; (\mathbf{x}_t, y_t)) \leq \xi, \tag{B.13}$$

The Lagrangian for this optimization problem is given by:

$$\mathcal{L}(\omega, \xi, \tau_t) = \frac{1}{2}\|\omega - \omega_t\|^2 + C_t\xi^2 + \tau_t \left[(\rho * \mathbb{I}_{(y_t=1)} + \mathbb{I}_{(y_t=-1)}) * \max(0, 1 - y_t(\omega \cdot \mathbf{x}_t)) - \xi \right], \tag{B.14}$$

Differentiating with respect to ω and ξ gives the optimality conditions:

$$\frac{\partial \mathcal{L}}{\partial \omega} = \omega - \omega_t - \tau_t \left(\rho * \mathbb{I}_{(y_t=1)} + \mathbb{I}_{(y_t=-1)} \right) y_t\mathbf{x}_t = 0 \tag{B.15}$$

$$\frac{\partial \mathcal{L}}{\partial \xi} = 2C_t\xi - \tau_t = 0 \implies \xi = \frac{\tau_t}{2C_t} \tag{B.16}$$

We substitute the expressions for ω and ξ into the Lagrangian, which results in a dual objective that is a function of τ_t. After simplification, the dual Lagrangian is:

$$\mathcal{L}(\tau_t) = -\frac{1}{2}\tau_t^2 \left(\rho * \mathbb{I}_{(y_t=1)} + \mathbb{I}_{(y_t=-1)} \right)^2 \|\mathbf{x}_t\|^2 - \frac{\tau_t^2}{4C_t} + \tau_t\ell^{II}(\omega_t; (\mathbf{x}_t, y_t)), \tag{B.17}$$

To find the optimal τ_t, we differentiate this dual form with respect to τ_t and set it to zero:

$$\frac{\partial \mathcal{L}}{\partial \tau_t} = -\tau_t \left(\rho * \mathbb{I}_{(y_t=1)} + \mathbb{I}_{(y_t=-1)} \right)^2 \|\mathbf{x}_t\|^2 - \frac{\tau_t}{2C_t} + \ell^{II}(\omega_t; (\mathbf{x}_t, y_t)) = 0, \tag{B.18}$$

Solving for τ_t provides the final update parameter:

$$\tau_t = \frac{\ell^{II}(\omega_t; (\mathbf{x}_t, y_t))}{\left(\rho * \mathbb{I}_{(y_t=1)} + \mathbb{I}_{(y_t=-1)} \right)^2 \|\mathbf{x}_t\|^2 + \frac{1}{2C_t}}. \tag{B.19}$$